MUSTAFA YALVAÇ

Electrical & Electronics Engineer — E-Drive Development & EV Quality

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PROFESSIONAL PROFILE

Electrical & Electronics Engineer with 5+ years in electric powertrain development and automotive quality, progressing from motor design engineer to R&D Lead of a multidisciplinary team. Delivered electromagnetic design, validation and industrialisation of traction machines for EV and industrial programmes including TOGG, ISUZU, KARSAN–ASELSAN and Nordex. Currently responsible for EV charging-efficiency validation, ECU/CAN diagnostics and new-model quality assurance on the IONIQ series at Hyundai Motor Manufacturing Turkey. Combines hands-on simulation and test capability with cross-functional leadership across R&D, production, suppliers and certification bodies.

CORE COMPETENCIES

Electric Machine Design	IPM, Synchronous Reluctance & Induction machines · electromagnetic topology · FEA optimisation
E-Mobility Systems	Electric Drive Unit (EDU) integration · traction motor sizing · inverter & battery supplier coordination
Test & Validation	Prototype validation · dyno & performance testing · charging efficiency · power measurement
Vehicle Diagnostics	CAN bus communication analysis · ECU fault diagnosis · Vector CANoe / CANalyzer
Quality & Compliance	Quality management systems · customer & certification audits · countermeasure and improvement systems
Leadership	R&D team building · project leadership · cross-functional coordination · technical training & documentation

PROFESSIONAL EXPERIENCE

Quality Engineer — EV Performance & Diagnostics — 2025 – Present

Hyundai Motor Manufacturing Turkey · Kocaeli, Türkiye

- Lead charging-efficiency testing and performance validation for the IONIQ EV series, verifying compliance with global EV standards and internal quality gates.
- Own vehicle-level electrical diagnostics: complex CAN communication analysis and ECU fault diagnosis with Vector CANoe and CANalyzer, closing issues raised in production and validation.
- Execute high-precision electrical measurements with Hioki power analysers to characterise charging performance, efficiency losses and power quality.
- Act as quality project leader for the BJ new-model programme, coordinating R&D and production teams through technical validation and launch readiness.
- Manage the GQMS (Global Quality Management System) improvement and suggestion pipeline, converting shop-floor findings into countermeasures that raise production efficiency and product quality.

AEMOT Electric Motors & Generators — 2021 – 2025

R&D Lead — 2025

- Built and managed a multidisciplinary R&D team of designers, simulation experts, test engineers and technical writers, setting priorities across the electric machine portfolio.
- Directed the complete development cycle — electromagnetic design, simulation, prototyping, validation testing and industrialisation — for motors and generators.
- Led the diesel-to-electric conversion of a forklift powertrain: specified and developed the dedicated traction motor and coordinated inverter and battery suppliers from concept to validated prototype.

Senior R&D Engineer

2023 – 2025

- Led electromagnetic design and optimisation of electric machines for industrial and e-mobility applications, from concept through design freeze.
- Supported international certification and customer audits by preparing technical documentation and leading on-site technical reviews.
- Extended scope into electric drive unit (EDU) architecture for electric vehicles while owning motor design, testing and commissioning.
- Delivered technical training on electric machine design and test methodology to universities, industry partners and internal engineering teams.

R&D Engineer

2021 – 2023

- Designed the electromagnetic topology of synchronous reluctance, induction and IPM machines using Motor-CAD, Altair Flux and Ansys Maxwell.
- Produced product datasheets and comprehensive test reports for the electric machine range.
- Supported production and test phases including coil winding, rotor–stator assembly and prototype validation.

Electrical & Electronics Engineering Intern

Jun – Aug 2017

Koçak Engineering · Aksaray, Türkiye

- Assisted in electrical project drafting and on-site implementation, including control panel setup and internal wiring systems.

KEY PROJECTS

TOGG EV Traction Motor	R&D electrical part responsible
ASELSAN – KARSAN Motor Programme	R&D electrical part responsible
ISUZU L7-Class Traction Motor	Project leader
Pilotcar Motor Project	Project leader
Nordex Wind Turbine Pitch & Yaw Motor	R&D electrical part responsible
Hyundai IONIQ 3 — EV Efficiency & Software Validation	Quality & diagnostics lead
Forklift Diesel-to-Electric Powertrain Conversion	Project leader — motor development & supplier integration

TECHNICAL SKILLS

Electric Machine Design	Ansys Maxwell, Altair Flux, Motor-CAD, Simcenter SPEED
Diagnostics & Measurement	Vector CANoe, Vector CANalyzer, Hioki Power Analyzer
CAD	SolidWorks, AutoCAD
Systems & Reporting	IFS ERP, Microsoft Office (advanced Excel reporting)

EDUCATION

B.Sc. Electrical and Electronics Engineering — Aksaray University 2015 – 2019

English Preparatory Year — Niğde University 2014 – 2015

LANGUAGES

Turkish	Native
English	Professional working proficiency